Automotive Manufacturing 🏭

Brownie Automotive Engineering Badge — Girls Build Bots

Presented by your robotics mentors!

One car… or a HUNDRED? 🚗 🚗 🚗

You designed a car. You engineered a car.

Now: how do you make hundreds — the same, fast, and well made?

The Assembly Line 🔗

Instead of **one person building a whole car**…
Each person does **one job** as the car moves past.
Henry Ford made this famous.

Round 1: Everyone Solo 🧍

Make one complete paper car **all by yourself**. Every single step.

⏱️ We’re timing you!

What did you notice? 🤔

- Was it **slow**? 🐢
- Did everyone's car come out a little **different**?

Now let's try it the factory way…

Round 2: Build the LINE! 🏭

Each Scout takes ONE station:

1. **Fold** 📄
2. **Wheels** 🛞
3. **Windows** 🪟
4. **Decorate** 🎨
5. **Inspect** 🔍

The cars move down the line. You do only your job — over and over!

The Quality Inspector 🔍

One Scout has the **checklist**.

Any car missing a part? **Send it back!**

Quality matters as much as speed.

Compare the rounds ⏱️

- Was the line **faster**?
- Were the cars more **alike**?
- What happened when one station got **backed up**?

The factory truth 

The line made **more cars, more alike, faster**…

…but **everyone had to work as a team**.

And **one slow station slowed everybody down.**

That’s real manufacturing!

Robot moment 

Our team splits build season into roles and stations — **programming, building, wiring** — so lots of work happens at once.

Just like a factory line!

You did it! 🎉

Today you: - ✅ Explained what an assembly line is and why factories use one - ✅ Worked a station as part of a manufacturing team - ✅ Compared building solo vs. building on a line - ✅ Said why quality checking matters

Let’s count our cars — that’s the TEAM’s total! 🙌

Keep going at home! 

- Run an assembly line at home: card-making, sandwich-making, anything!
- Spot the “stations” next time you see people working together
- Ask your troop leader about your next STEM badge!

📣 Share Your Impact

Troop leaders & robotics teams — if you used this curriculum, please:

1. **Credit us** — the curriculum comes from **Girls Build Bots**
2. **Tag us** — **@girlsbuildbots** on Instagram
3. **Report your results** — email **girlsbuildbots@gmail.com** with how many girls took part, how many badges were earned, and how it went

This is how the program grows — and how the next troop and team get better materials.